

Labour Market

MRP theory
and
Perfect competition

Any

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Labour Market



Demand
for Labour

wages ↑ Qty d_L ↓
wages ↓ Qty d_L ↑

Negative
relationship

Supply
of Labour

wages ↑ Qty s_L ↑
wages ↓ Qty s_L ↓

Positive
relationship

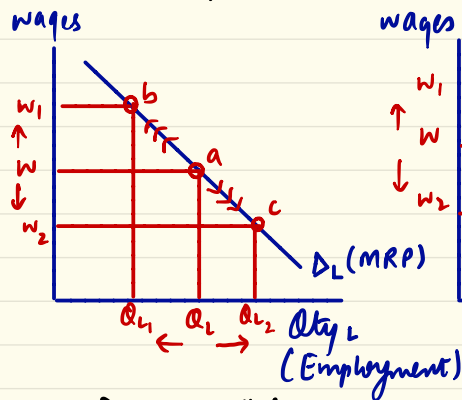


Diagram #1

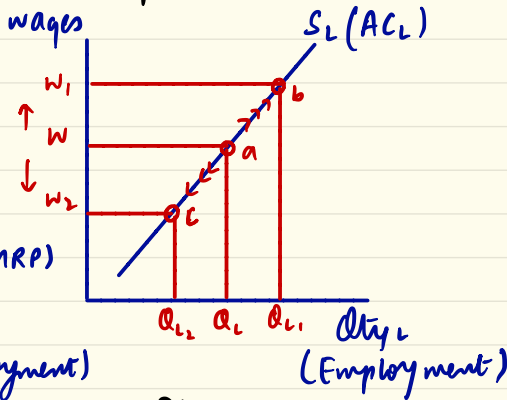


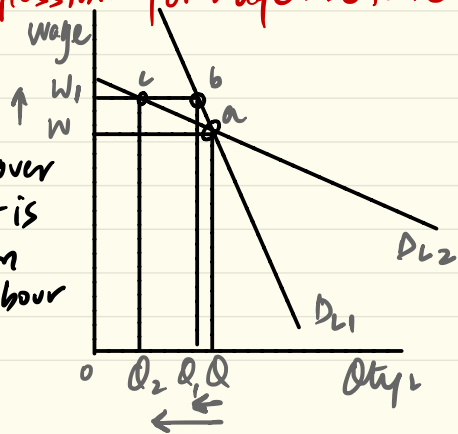
Diagram #2

Note:- An increase in wage rate, increases cost of production thus results in contraction in quantity demanded for labours as shown in diagram #1, a downward sloping demand for labour diagram representing negative relationship. Further in this we will identify this being MRP (Marginal revenue productivity).

Note:- In diagram #2, an increase in wage due to increase in demand for labour results in shortage, which increases wages, thus extension in quantity supplied of workers, proving positive relationship thus an upward sloping supply of labour. It is also known as average cost of labour, minimum wage the worker is ready to accept

Elasticity of demand for labour:

- * If labour cost in a firm are high relative to total costs of production, the elasticity of demand will be high because any given increase in wages will have a significant impact on total costs. For e.g. if a firm's total costs are \$1 million, of which 90% or \$900,000 are labour costs, then a 10% increase in wages will raise cost substantially by \$90,000. It may prove difficult for the firm to absorb this cost increase by reducing profits or increasing prices. In this case it will have to shed proportionately more labour. However, if labour costs constituted only 10% of total costs then total costs will rise by only \$100,000, which the firm might find easier to pass on to consumers or absorb by making savings elsewhere.
- * If labour can easily be replaced by other factors, particularly capital, then the demand for labour is elastic. Such factor substitution occurs with unskilled labours.
- * If the PED of final product is high then it will not be possible for wage rise to be passed on to consumers, so demand for labour will also be elastic. This is likely to be the case in competitive product market which means that consumers can switch easily.
- * The elasticity of demand for labour is likely to increase over time as firms have longer to find substitutes for labour. It is also the case that in some countries government legislation and employment law make it difficult for firms to shed labour in very short period of time.



Elasticity of Supply of labour:-

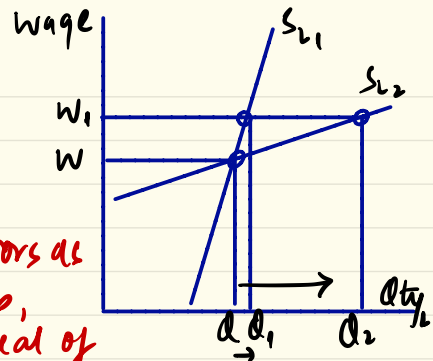
* If labour is highly mobile, both occupationally and geographically, the elasticity will be high too.

The occupational mobility of labour depends on such factors as the level of education, training and skills required by the job, together with the length of training. It requires a great deal of education and skill as well as many years of training to become barrister or cardiac surgeon and so supply of labour is inelastic. If job requires little skill/training, such as stacking shelves in a supermarket, then supply of labour is elastic.

Geographical mobility is influenced by factors such as family ties and the nature of the housing market. In many countries, house prices are generally much higher in some areas and this can act as a disincentive to take up jobs in different regions even if they are available, this reduces elasticity of supply.

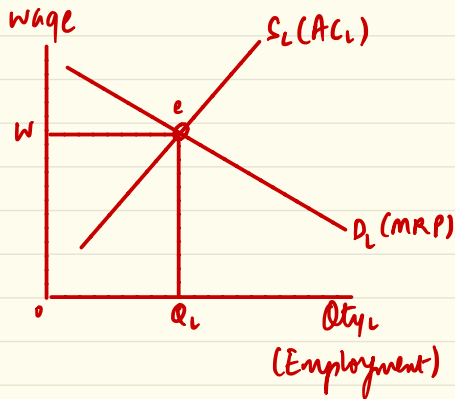
* Unemployment in the economy, if there are high levels of unemployment in an economy, then there will be large pool of available labour and supply should be more elastic, but again mainly for industries requiring unskilled labour.

* Time period, the longer the time period, the more elastic the labour supply is likely to become. Even in professions such as law and medicine, given a sufficiently long period of time, more barristers and cardiac surgeons can be trained to the required level.

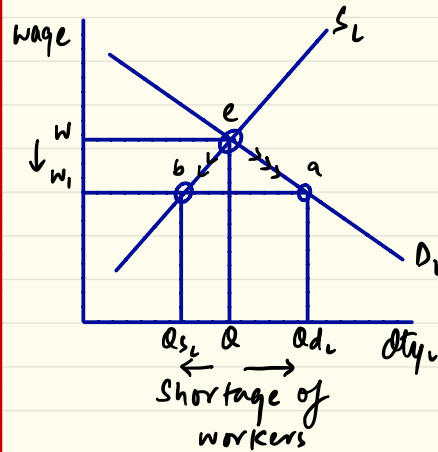


Equilibrium under Labour market

$$D_L = S_L$$

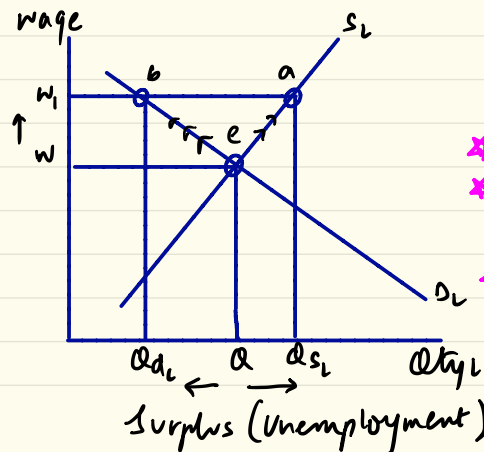


Dis-equilibrium ($D_L \neq S_L$)



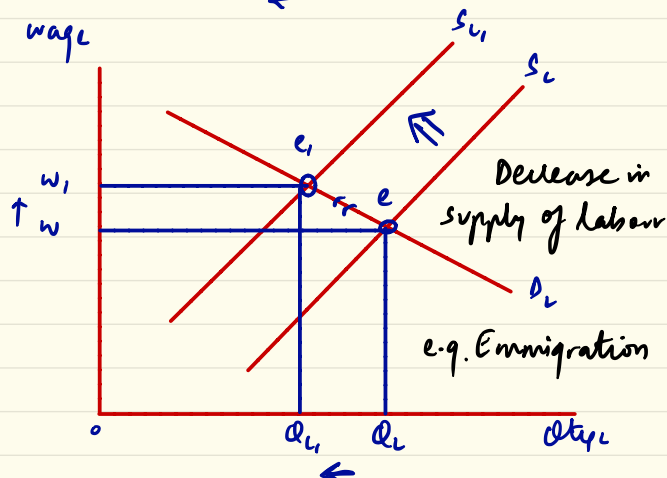
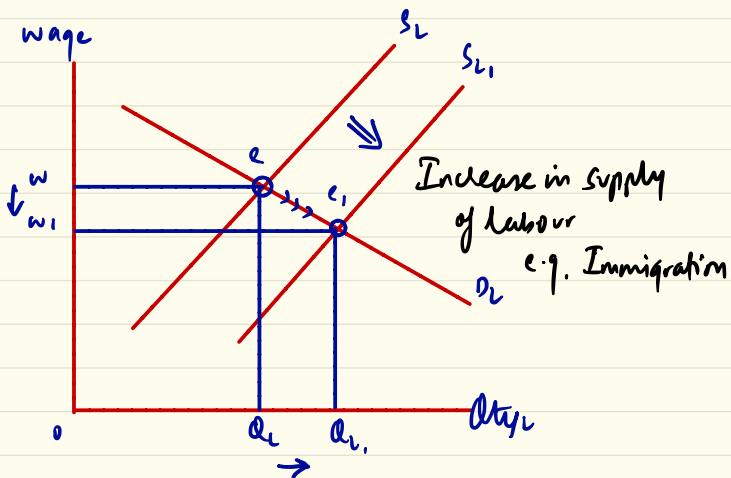
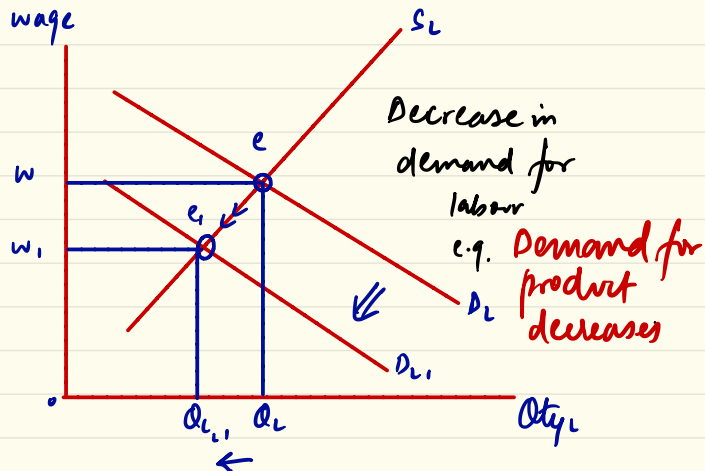
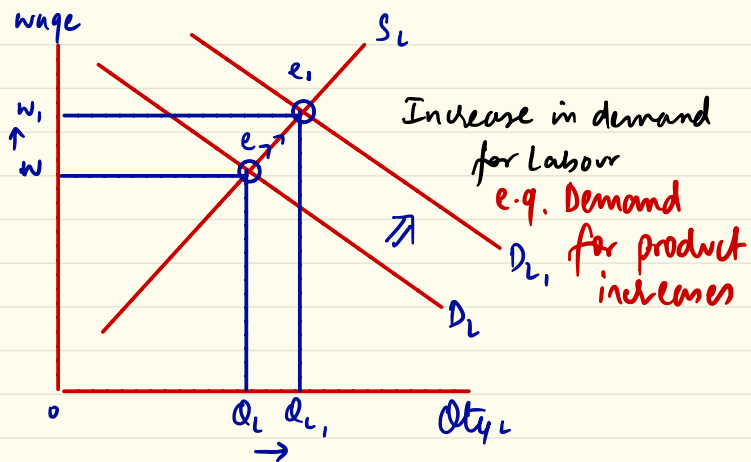
wages ↓

- ★ Qty demand_L ↑
- ★ Qty supply_L ↓
- ★ $Q_{d_L} > Q_{s_L} \therefore$ Shortage



wages ↑

- ★ Qty supply_L ↑
- ★ Qty demand_L ↓
- ★ $Q_{s_L} > Q_{d_L} \therefore$ Surplus



Perfect competition under Labour market (Remember Market Structures)

Market structures

vs

Labour market

- * 1000 of producers
- * same product
- * same price
- * price taker
- * $AR = MR$
- * No Price discrimination

Price	Qty	TR	AR	MR
5	1	5	5	5
5	2	10	5	5
5	3	15	5	5
5	4	20	5	5
5	5	25	5	5
5	6	30	5	5

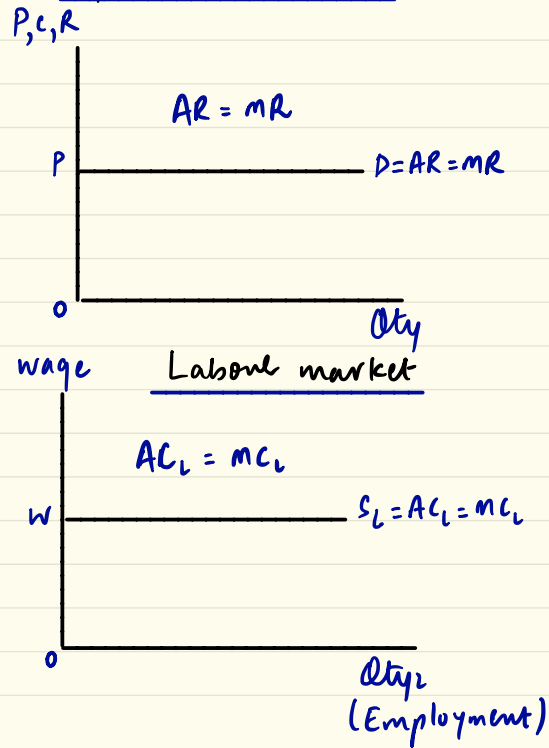
- * 1000 of workers
- * ability and productivity same
- * same wage
- * wage taker
- * $AC_L = MC_L$
- * No wage discrimination

wages	Qty _L	TC _L	AC _L	MC _L
5	1	5	5	5
5	2	10	5	5
5	3	15	5	5
5	4	20	5	5
5	5	25	5	5
5	6	30	5	5

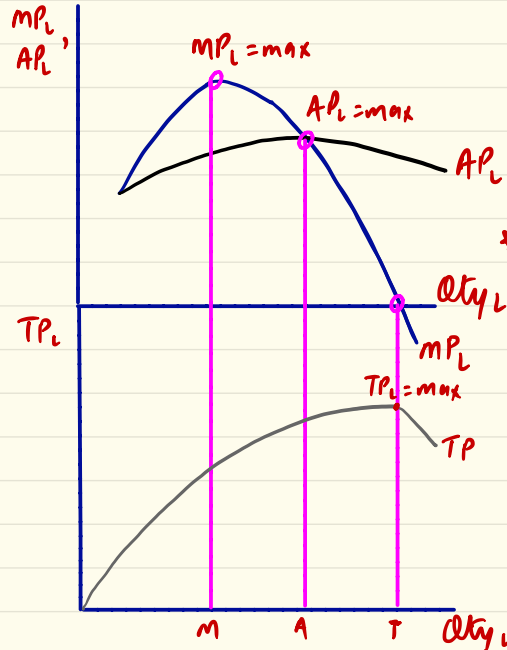
Remember MR, DARP

Note:- Under perfect competition, there are 1000s of workers ready to supply them at a given wage, thus average cost of labour (in other words wage) is equal to marginal cost of labour (cost of hiring additional labour), $AC_L = MC_L$

Market structures



Remember MAT theory



* Marginal product of labour, additional labour contributing to the additional output. $\frac{\Delta TP}{\Delta L}$

* Average product of labour is labour productivity that is a labour contributing to the total output $\frac{\text{Total output}}{\text{No. of Labours}}$

Law of diminishing returns!

Introducing Marginal Revenue Productivity (MRP Theory)

The marginal revenue which the firm gets from employing one more worker is called MRP. It is the value of marginal product of labour.

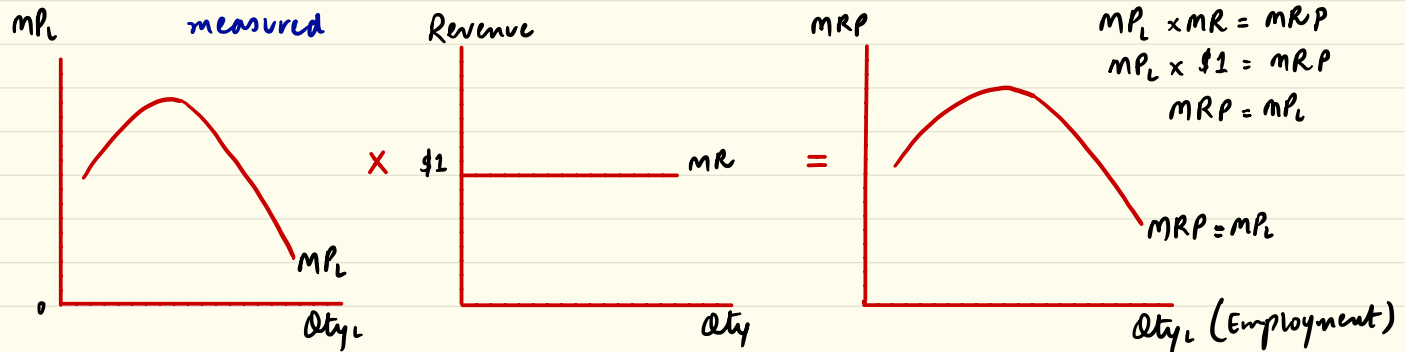
$$\text{MRP} = \text{MP}_L \times \text{MR} \Rightarrow \text{Marginal revenue:- Revenue received by selling additional unit.}$$

$\downarrow \qquad \downarrow$
MR is constant under perfect competition. (Review last slide)

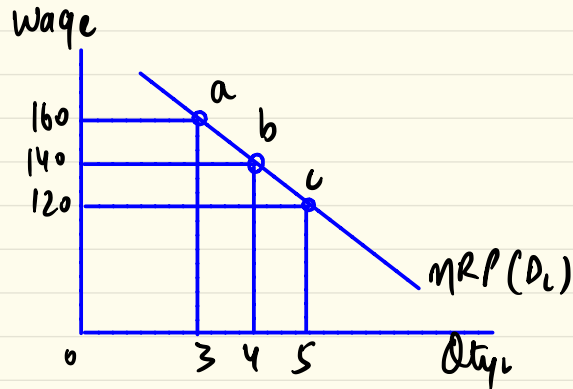
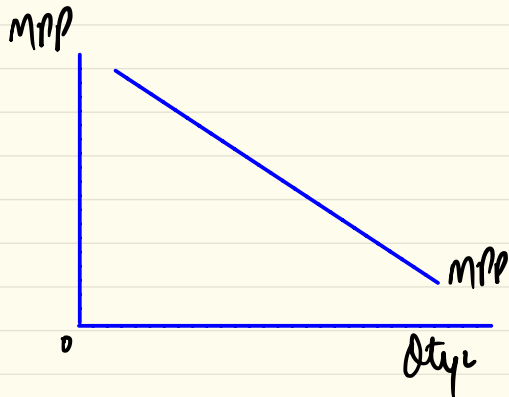
* MR
* MP_L

MPP:- Marginal physical product
 $\downarrow$
physical productivity of each labour can be calculated and measured

Note:- The MP_L curve follows the law of diminishing returns (Review last slide) i.e. MP_L decreases due to increasing returns and falls due to decreasing returns. The MRP curve is the product of MP_L and a constant MR (e.g. \$1), so the shape of MRP resembles the shape of MP_L curve.



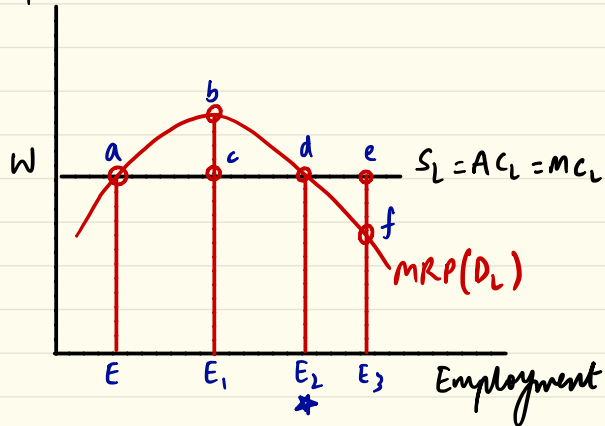
Number employed	mpp or mPL	Price = MR \$	MRP = mpp x MR(p)	Wage
1	20	10	200	160
2	18	10	180	160
3	16	10	160	160
4	14	10	140	160
5	12	10	120	160
6	10	10	100	160



Profit maximizing level of employment vs Profit maximizing level of output

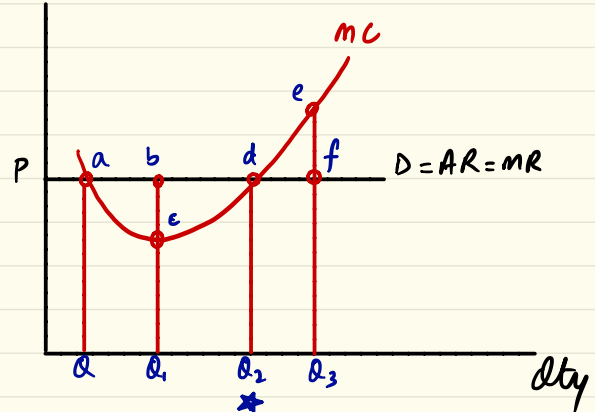
$$MRP = MC_L$$

wage



$$MR = MC$$

P, C, R



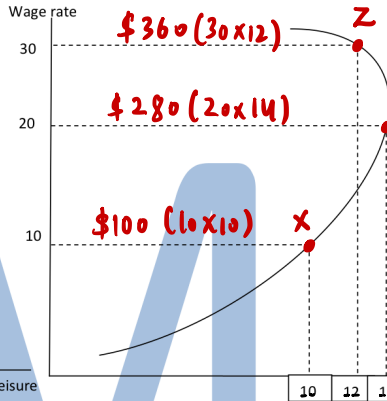
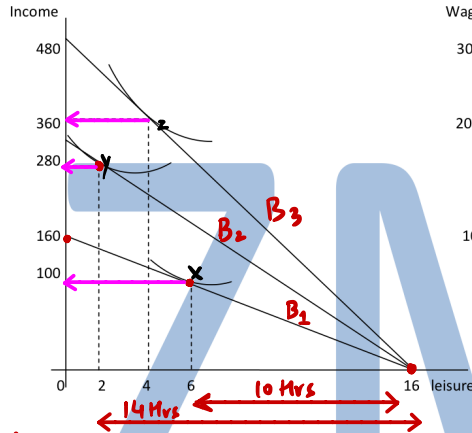
Note:- At employment 'E' $MRP = MC_L$ but that is not profit maximizing level of employment, as there is still additional revenue productivity (MRP) which can be reviewed. At 'E₁', $MRP > MC_L$ but even a greater employment, still contributing to the revenue, additional is falling, but total is still rising at a decreasing rate. At 'E₂' it is the profit maximizing level of employment as employment greater than this level, worker is contributing more to the additional cost than to the additional productivity.

The individual supply curve (income - leisure trade off)

The individual supply curve for labour can be explained with the help of income-leisure trade off. A rise in wage rate implies a change in relative price of income and leisure. Income becomes cheaper relative to leisure, each hour worked earns more income than before or leisure becomes more expensive, since each hour of leisure consumed is not at the expense of more income forgone.

This change in relative prices has both income and substitution effects. The substitution effect leads the individual to earn more income and less of more expensive leisure. But after achieving a certain level of income, a further increase in the wage rate may persuade the labour to enjoy greater income and greater leisure time also. This is called the income effect of the wage increase and is achieved when the increase in wage rate leads to fall in the number of hours worked, provided that the rise in wage rate proportionately greater than the reduction in the number of hours of work.

Income
and
Substitution
effect!



Backward
Bending
Supply
Curve!

- * Assume that an individual can choose the number of hours to work and has 16 hours a day to divide between work and leisure. In the diagram, with an hourly wage rate of \$10, budget line B_1 shows all the possible combinations of daily income and leisure hours. For e.g. at point X the individual has an income of \$100 by working 10 hours and taking 6 hours of leisure.
- * At an hourly rate of \$20 the budget line become B_2 and the labor increases the hours of work or substituted the hours of leisure for income. He maximizes utility at point Y by working 14 hours and earns an income of \$280. Thus the higher wages have encouraged the individual to work more number of hours.
- * As the wage rate increases to \$30 per hour the budget line moves to B_3 where the labor works for 12 hours and enjoys greater income of \$360 and greater leisure time or less hours of work. The income effect leads the individual to earn more income and more leisure provided that the reduced number of hours of work is proportionately less than the increased wage rate.

Substitution
effect

wage $\uparrow$ Hrs $\uparrow$
leisure $\downarrow$

Income
effect!

Because of the substitution effect the labor supply curve will be upward sloping and because of income effect the supply curve will be downward sloping. So up to some wage rate the number of hours of work and wage rate will positively related and after that the curve will bend backwards i.e. wage rate and hours worked will be negatively related.

Questions

- Q1 Explain labour market under perfect competition, using diagram and table? (8)
- Q2 Explain, using diagrams, marginal revenue productivity theory under perfect competition? (8)
- Q3 Explain, using a diagram, condition of profit maximizing level of employment under perfect competition? (8)
- Q4 Explain the factors affecting elasticity of demand of labour? (6) → 2 factors
(Diagram)
- Q5 Explain the factors affecting elasticity of supply of labour? (6) → 2 factors
(Diagram)
- Q6 Explain, using diagram(s), the concept of backward bending supply curves? (12)

Define

Perfectly elastic supply
↑

Derivation of MRP

Define MRP + Formula!

$$MRP = MC_L$$